Optimization - Revenue and Cost Questions

1. A railway company offers excursions at $120/person to groups of up to 100 people. If the size of the group is greater than 100, the company will reduce the price of every ticket by 50 cents for each person in excess of 100. What size group would produce the greatest revenue?

2. CAP wishes to sell teddy bears to raise money. If CAP charges $20, they can sell 140 teddy bears per week. For every $2 increase in price they sell 5 less teddy bears. What price should CAP charge to maximize revenue?

3. A telecommunications company is laying fiber optic cable for a prison located on an island that is 100 m from the mainland. The company has set up a building on the mainland that is 1.2 km from the point on the shore that is closest to the island. If it costs $40/m to lay the cable underground and $80/m to lay the cable under water, what is the least expensive way to lay the cable?

4. An open topped storage box with a square base is to have a capacity of $5m^3$. Material for the sides costs $\$1.60/m^2$ while the cost for the bottom is $\$2/m^2$. Find the dimensions that will minimize the cost of the material. What is the minimum cost?

5. A store can sell 320 shirts a week when they are priced at \$24. For every \$2 decrease in price, they can sell 10 more shirts. If the store's fixed cost for the shirts is \$300 weekly plus the shirts cost them \$4 each:

a) what price should they charge to maximize profit?

b) what is the maximum profit?

6. A juice can in the shape of a right circular cylinder has a **fixed** capacity. If the material used for the sides of the can costs \$0.50/cm^2 and the material for the top and the bottom costs \$0.25/cm^2, determine the height and the radius of the cylinder that result in a minimum cost.